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TOPIC: GRAPHS

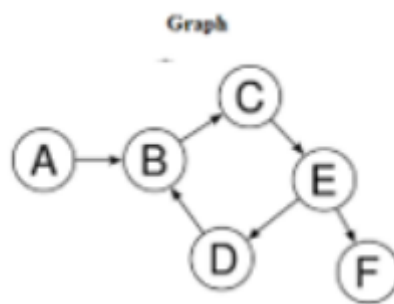
SUBMITTED TO: MA'AM NEHAN

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LAB TASK 1:

- Make Adjacency Matrix of above graphs.
- Make Adjacency list of above graphs.
- Perform BFS.
- Perform DFS.

Figure 1: Directed Graph (Digraph)



Adjacency Matrix

	A	B	C	D	E	F
A	0	1	0	0	0	0
B	0	0	1	0	0	0
C	0	0	0	0	1	0
D	0	0	0	0	1	0
E	0	0	0	1	0	1
F	0	0	0	0	0	0

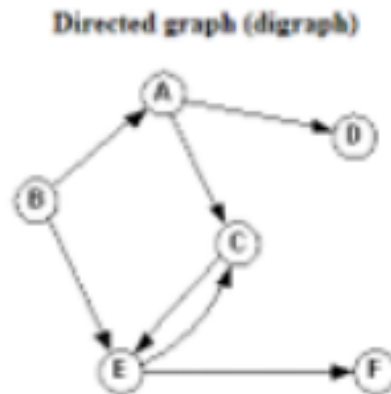
Adjacency List

- A → B
- B → C
- C → E
- D → B
- E → D, F
- F →

BFS (start A): A → B → D → C → E → F

DFS (start A): $A \rightarrow B \rightarrow C \rightarrow E \rightarrow F \rightarrow D$

Figure 2: Directed Graph (Digraph)



Adjacency Matrix

	A	B	C	D	E	F
A	0	0	1	1	0	0
B	1	0	0	0	1	0
C	0	0	0	0	1	0
D	0	0	0	0	0	0
E	0	0	1	0	0	1
F	0	0	0	0	0	0

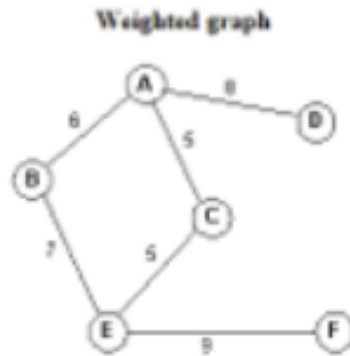
Adjacency List

- $A \rightarrow C, D$
- $B \rightarrow A, E$
- $C \rightarrow E$
- $D \rightarrow$
- $E \rightarrow C, F$
- $F \rightarrow$

BFS (start B): $B \rightarrow A \rightarrow E \rightarrow D \rightarrow C \rightarrow F$

DFS (start B): $B \rightarrow A \rightarrow D \rightarrow C \rightarrow E \rightarrow F$

Figure 3: Undirected, Weighted Graph



Adjacency Matrix

	A	B	C	D	E	F
A	0	6	5	8	0	0
B	6	0	0	0	7	0
C	5	0	0	0	5	0
D	8	0	0	0	0	0
E	0	7	5	1	0	9
F	0	0	0	0	9	0

Adjacency List

- A → B, C, D
- B → A, E
- C → A, E
- D → A
- E → B, C, F

BFS (start A): A → B → D → C → E → F

DFS (start A): A → B → E → F → C → D

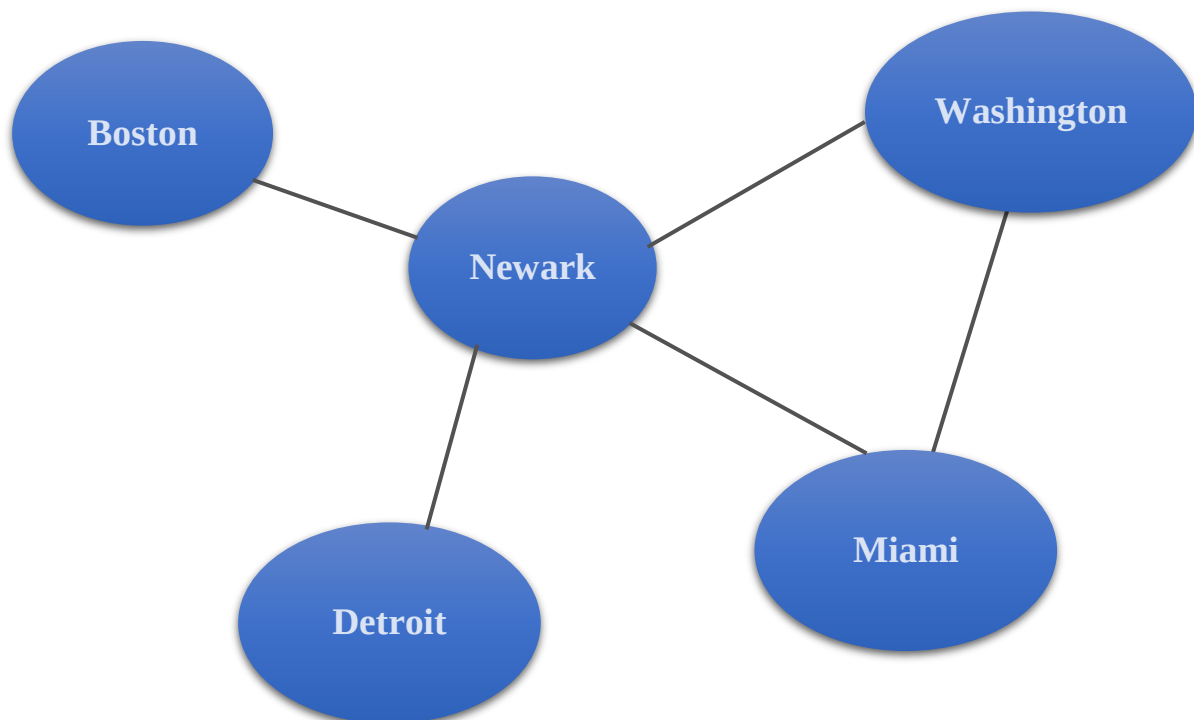
Lab Task 2:

Draw a graph modes, starting the type of graph used, to represent airline routes where every day there are four flights from Boston to Newark, two flights from Newark to Boston, three flights from Newark to Miami, two flights from Miami to Newark, one flight from Newark to Detroit, two flights from Detroit to Newark, three flights from Newark to Washington, two flights from Washington to Newark, and one flight from Washington to Miami, with

- (1) An edge between vertices representing cities that have a flight between them (in either direction)
- (2) An edge between vertices representing cities for each flight that operates between them (in either direction)
- (3) An edge between vertices representing cities for each flight that operates between them (in either direction) plus a loop for a special sightseeing trip that takes off and lands in Miami

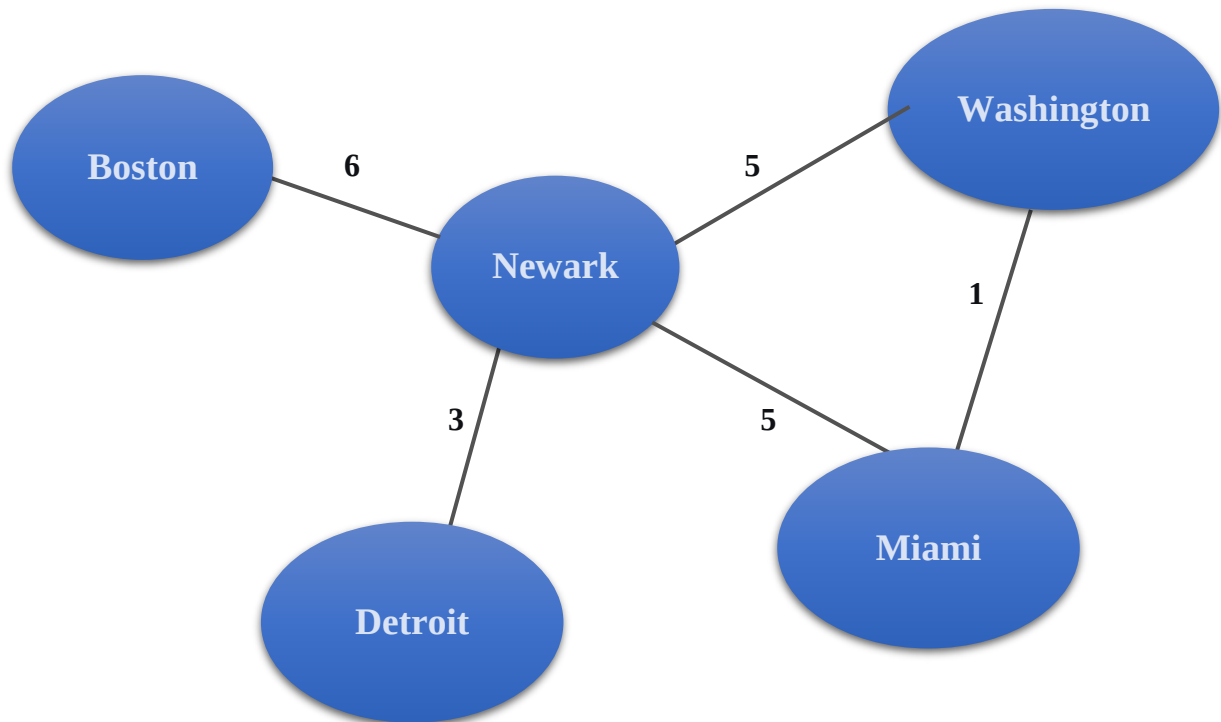
Graph Model (Case 1: Simple Connectivity)

- Undirected simple graph (no multiple edges, no loops)



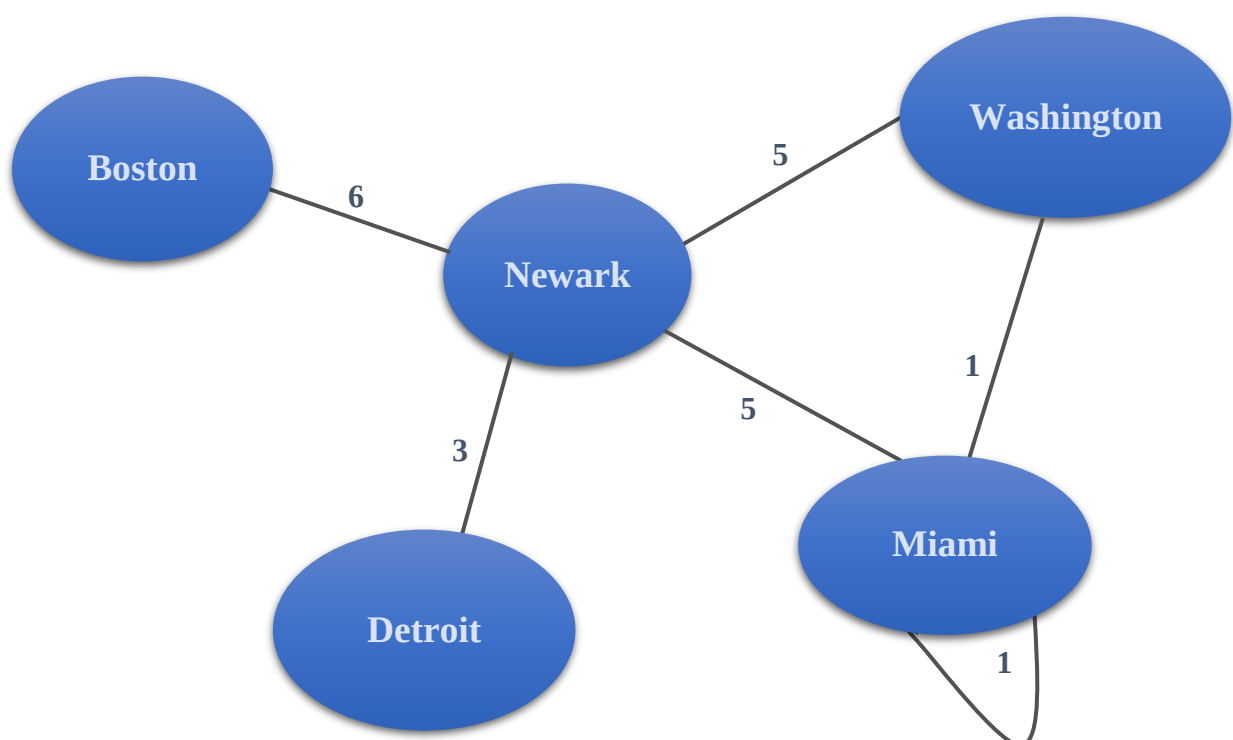
Graph Model (Case 2: Flight Frequency as Multiple Edges)

- Undirected multigraph (multiple edges allowed, no loops)



Graph Model (Case 3: Weighted Graph with Loop)

- Undirected graph (multiple edges and loops allowed).



Lab Task 3

Describe a graph model that represents whether each person at a class knows the name of each other person at the class.

Should the edges be directed or undirected?

Should multiple edges be allowed?

Should loops be allowed?

Answer:

1. Graph Type

- **Vertices:** Each person in the class.
- **Edges:** Represent “knows the name of.”

Should edges be directed or undirected?

- If *knowing is mutual* (A knows B, B knows A), then **undirected graph**.
- If *knowing is one-way* (A knows B, but B doesn’t know A), then **directed graph**.

In most classroom settings, we assume **undirected** because knowing names is usually mutual.

Should multiple edges be allowed?

- **No.** One edge is enough to represent “knows.”
- Multiple edges would be redundant.

Should loops be allowed?

- **No.** A person doesn’t need an edge to themselves (self-knowledge is trivial).

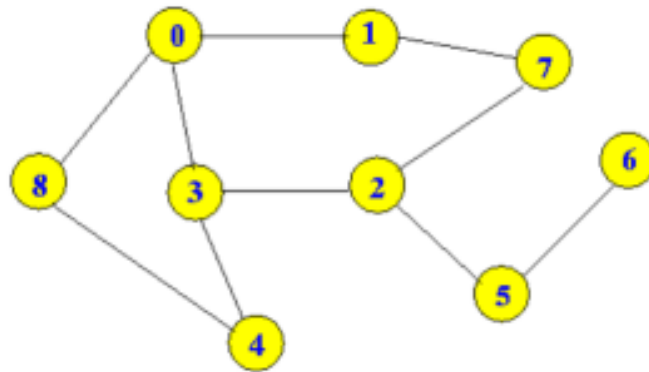
2. Graph Model

- **Type:** Undirected simple graph.
 - **Vertices:** Students.
 - **Edges:** Connect two students if they know each other’s names.
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Lab Task 4:

- Perform BFS
- Perform DFS

Given Graph:



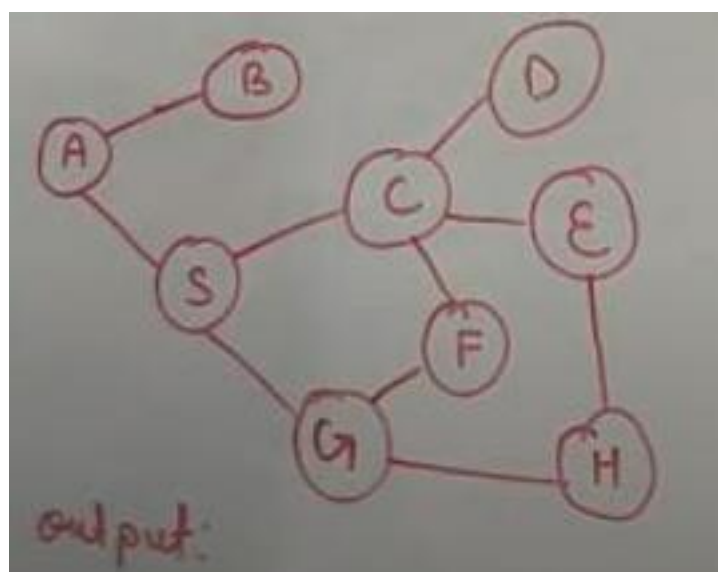
BFS Order (Starts from 0):

$0 \rightarrow 8 \rightarrow 3 \rightarrow 1 \rightarrow 2 \rightarrow 4 \rightarrow 7 \rightarrow 5 \rightarrow 6$

DFS Order (Starts from 0):

$0 \rightarrow 1 \rightarrow 7 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 8 \rightarrow 5 \rightarrow 6$

Given Graph:



BFS Order (Starts from A):

$A \rightarrow B \rightarrow S \rightarrow C \rightarrow G \rightarrow D \rightarrow F \rightarrow E \rightarrow H$

DFS Order (Starts from A):

$A \rightarrow B \rightarrow S \rightarrow G \rightarrow H \rightarrow E \rightarrow C \rightarrow D \rightarrow F$
